

DINUKA WITHANAGE

Computer Science and Engineering Undergraduate

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OBJECTIVE

Computer Science and Engineering undergraduate at the University of Moratuwa with interests in software engineering, artificial intelligence, machine learning and data science. Experienced in developing full-stack applications and machine learning solutions through academic projects. Interested in applying strong problem-solving and software development skills to build scalable, practical, and impactful technology solutions.

EDUCATION

B.Sc. Engineering (Hons.) in Computer Science and Engineering 2024 – Present
University of Moratuwa

- Specialized in Data Science Engineering Stream
- CGPA: 3.22 / 4.00** (1–4 semesters)

G.C.E. Advanced Level – Physical Science Stream 2022(2023)
Dharmasoka College, Ambalangoda

- Results: Combined Mathematics (A), Physics (A), Chemistry (A)
- Common General Test: 72/100
- Z-Score: 2.6801**

PROJECTS

BrightBuy – Full-Stack E-Commerce Application Aug 2025 – Nov 2025
React, Node.js, Express.js, SQLite, Tailwind CSS

- Developed a full-stack e-commerce application with responsive user interfaces and RESTful backend services.
- Built frontend components using React and Tailwind CSS and implemented backend functionality with Node.js and Express.js.
- Integrated SQLite for persistent application data management.

WildGuard – Wildlife Education & Recognition Platform Dec 2025 – Jan 2026
React, Node.js, Express.js, SQL, REST APIs, AI Image Recognition

- Developed a full-stack wildlife education platform with species browsing, quizzes, events, and user dashboards.
- Integrated AI-powered image recognition and interactive species mapping for wildlife identification and exploration.
- Implemented secure authentication, RESTful APIs, SQL data management, and quiz leaderboards.

Zeta-26 Relic Ring Protocol Simulator – Launch26 2026
JavaScript, HTML, CSS, Dijkstra's Algorithm, JSON, Docker, Nginx

- Developed an interactive network routing simulator with real-time packet transmission visualization.
- Implemented Dijkstra's algorithm for minimum-latency routing and dynamic rerouting.
- Built latency analysis and Base-N data translation with Docker and Nginx deployment.

Nano Processor – Versions I & II Mar 2025 – May 2025
VHDL, Digital Logic Design, FPGA

- Designed and implemented a custom nano processor using VHDL as part of digital systems coursework.
- Developed core processor components including the ALU, register file, control unit, and instruction execution pipeline.

Voice Translator 2026
Speech Recognition, Translation APIs

- Developed a speech-based language translation application that converts spoken input into translated text.
- Integrated speech-recognition and translation APIs to process voice input and generate translated output.

Energy Management System Feb 2026 – May 2026
Docker, Kubernetes, AWS, Prometheus, Grafana, Terraform

- Designed the monitoring and observability module for a cloud-native microservices-based energy management system.

- Integrated Prometheus and Grafana for service monitoring and visualization, with Docker and Kubernetes for containerized deployment.
- Applied Terraform and AWS technologies for infrastructure provisioning and cloud deployment.

Privacy-Preserving Multi-Agent AI System for Research Document Review

Ongoing

Multi-Agent AI, LLMs, Privacy-Preserving AI

- Developing a privacy-preserving multi-agent AI system for automated research document analysis and review.
- Designing specialized AI agents for collaborative document processing while maintaining data privacy.
- Exploring secure LLM workflows for agent coordination and structured research feedback.

Trustworthy Deep Learning for Chest X-ray Disease Detection

Ongoing

Python, PyTorch, DenseNet121, CNN, Grad-CAM, Deep Learning

- Developing a model-agnostic shortcut-suppression module for trustworthy chest X-ray disease classification.
- Implementing DenseNet121 with transfer learning, augmentation, and hyperparameter tuning as the baseline model.
- Evaluating performance, Grad-CAM explainability, calibration, and cross-hospital robustness.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, Dart

Frontend: React, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

AI / Data Science: Pandas, NumPy, Scikit-learn, Feature Engineering, Exploratory Data Analysis

Cloud / DevOps: Docker, Kubernetes, AWS, Terraform, Prometheus, Grafana, Git

Tools & Technologies: VS Code, Ubuntu, Flutter, Figma

Design & Media: Adobe Photoshop, Adobe Illustrator, Premiere Pro, CorelDRAW, Canva

AWARDS & ACHIEVEMENTS

- 3rd Place – HaXtreme 2025 Hackathon
- 3rd Place – Launch'26
- Finalist – Codemania 6.0, BitCode 6.0, CryptX 2.0
- Mahapola Higher Education Scholarship

LEADERSHIP & EXPERIENCE

Finance Committee Co-Lead 2026 - Present

IEEE CIS Student Branch Chapter, University of Moratuwa

Finance Committee Member 2026 - Present

IEEE IAS Student Branch Chapter, University of Moratuwa

Finance Committee Member 2025 - Present

CSE Student Society, Department of Computer Science and Engineering, University of Moratuwa

Events Committee Member 2026 - Present

IEEE Computer Society, University of Moratuwa

VOLUNTEERING & COMMUNITY

Finance Committee Lead – OctWave 3.0 Jul 2026

IEEE IAS Student Branch Chapter, University of Moratuwa

Finance Committee Lead – MindModel 2026

IEEE CIS Sri Lanka Chapter

Finance Committee Member – MoraForesight 4.0 Jul 2026

IEEE Student Branch, University of Moratuwa

Logistics Committee Member – CSE Career Fair 2026 Jan 2026

Department of Computer Science and Engineering, University of Moratuwa

Events Committee Member – ADScAI 2026 Apr 2026

Department of Computer Science and Engineering, University of Moratuwa

Program Committee Member – CSE Hit the Ground <i>Department of Computer Science and Engineering, University of Moratuwa</i>	Jan 2026
Organizing Committee Member – IEEE GAMMEDDATA API 4.0 <i>IEEE IAS Student Branch Chapter</i>	Sep 2025
Partnership Pitching Committee Member – Binara Padura 4.0 <i>Rotaract Club, University of Moratuwa</i>	Sep 2025

CERTIFICATIONS

- Mobile Application Development Course – IDET (*Ongoing*)
- Modern AI, Machine Learning & Deep Learning Course (*Ongoing*)
- Introduction to Machine Learning – Kaggle (2025)
- Intermediate Machine Learning – Kaggle (2025)
- Feature Engineering – Kaggle (2025)
- Data Visualization with Python – Kaggle (2025)
- Data Cleaning – Kaggle (2025)
- Problem Solving (Solve) – HackerRank (2025)
- Python for Beginners – Department of Computer Science & Engineering, University of Moratuwa (2023)
- English Language Proficiency Course – Southern English Training Academy (SETA)

INTERESTS

Artificial Intelligence, Software Engineering, Startups, UI/UX Design, Competitive Programming, Content Creation

REFERENCES

Prof. Dulani Meedeniya

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Semester 5 Coordinator

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